Partisan Vision? Partisan Bias in Simple Visual Evaluations*

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Abstract

In the current era of partisan polarization, partisanship strongly colors partisans' worldviews. But does it cause partisans to *see* different things? In one experiment, respondents count errors in a short passage; in another, they count poorly parked cars in a photograph. We vary the party attributed to the author or drivers and replicate the writing experiment on Lucid. Pooling the experiments yields an effect of 0.12 standard deviations (SE = 0.04) toward more favorable judgments of one's own party. The clearest individual effect is in parking: Democratic respondents count 2.63 more poorly parked cars under the out-party cue (SE = 0.73); the corresponding difference among Republican respondents is 0.34 (SE = 0.95). Writing evidence is weaker and sensitive to extreme responses in Lucid. In a separate MTurk survey, Republicans count more masked people than Democrats in a video of Trump supporters when extreme responses are capped. The experiments suggest a modest average partisan bias that varies across tasks; the mask-count evidence is less stable.

*Replication materials are available at https://github.com/soodoku/partisan_vision.

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Partisans are increasingly polarized (Iyengar, Sood and Lelkes 2012), with partisan cleavages outstripping some of the longer-standing racial cleavages (Iyengar and Westwood 2015). In this paper, we explore whether polarization affects how partisans "see." We test the hypothesis with simple visual evaluative tasks. In particular, we field two survey experiments and a survey. Uncapped writing estimates favor the respondent's own party, but the Lucid Republican estimate is sensitive to extreme responses. The pooled experimental evidence points toward a modest partisan bias, with the clearest individual effect on Democrats' counts of poorly parked cars.

Group loyalties can shape how people judge the same scene. In the classic study of a contentious football game, Princeton and Dartmouth students differed in the infractions they attributed to the teams (Hastorf and Cantril 1954). In a political setting, changing the purported cause of a protest changed judgments of whether the same demonstrators were obstructing or threatening pedestrians (Kahan et al. 2012). Related experiments find that motivations influence interpretations of ambiguous visual stimuli (Balcetis and Dunning 2006) and that political agreement affects which photographs people regard as representative of a candidate's skin tone (Caruso, Mead and Balcetis 2009).

These findings fit a broader literature on motivated political reasoning. People evaluate congenial arguments more favorably and scrutinize contrary arguments more closely (Taber and Lodge 2006). Party identification can organize attention, memory, and evaluation around defending one's political group (Van Bavel and Pereira 2018). A meta-analysis of experiments that hold information constant while varying its political implications finds bias among both liberals and conservatives (Ditto et al. 2019).

But partisan differences in survey answers need not be large or uniform. Accuracy incentives reduce partisan gaps in answers to factual questions (Bullock et al. 2015; Prior, Sood and Khanna 2015), and vague response options can inflate their size (Roush and Sood 2023). Even a straightforward comparison of inauguration photographs can elicit answers

favoring a respondent's preferred candidate (Schaffner and Luks 2018). Our experiments ask whether party labels affect how many errors people find in a passage or how many cars they judge to be poorly parked. Respondents assess the material in front of them and give a count, rather than recall a political fact or choose among verbal descriptions of it.

1 Data and Research Design

To assess how partisans *see*, we fielded two survey experiments with respondents recruited by YouGov as part of a Cooperative Congressional Election Study (CCES) module. In the first experiment, we presented people with a short passage and asked them to count the number of errors in it. We manipulated the perceived party of the person writing the text (see Figure SI 1.1). In the second experiment, we showed people a photo of a parking lot and asked them to estimate the number of poorly parked cars. We manipulated the perceived party of the people who parked the cars by changing the caption indicating where the photo was taken (see Figure SI 1.2). We replicated the first experiment with a convenience sample recruited through Lucid, using a shorter passage. A validation study found that a Lucid sample came closer to national demographic benchmarks than an MTurk sample and reproduced several established experimental effects (Coppock and McClellan 2019).

We complemented the survey experiments with a partisan evaluative task in a survey. In a survey conducted on MTurk, we asked respondents to watch a short video showing people wearing Trump clothing and estimate how many people in the video were wearing masks. In particular, our directions were as follows: "Please watch the following short (10-second) video. You will be asked a question about it on the next screen. ... How many people in the video were wearing masks?"

We analyze the unmatched CCES sample without survey weights and report results from the matched, weighted sample in the replication materials. The writing task appeared

before the election, and the parking task afterward. The main results focus on Democrats and Republicans, including leaners. The appendix reports descriptive results for all groups, including independents. For Lucid, we exclude previews and retain respondents who consented and passed the attention check. For MTurk, we pool the two survey waves and retain respondents who said they gave humorous or insincere answers never or rarely. The original instruments, party coding, sample counts, and sensitivity analyses are included in the replication materials.

Figure 1 presents scaled means alongside standardized out-party minus in-party differences (Hedges' g, the small-sample-corrected version of Cohen's d). The pooled estimate combines the six study-by-party contrasts from the writing and parking experiments. We estimate their covariance by bootstrapping respondents within study and party, keeping each CCES respondent's writing and parking outcomes together. This accounts for clustering at the respondent level. The syntheses and the tests of moderation by partisan strength are exploratory; the appendix describes their construction. All intervals are 95% confidence intervals. We examine uncapped and capped responses, the influence of individual observations, and differences across the two MTurk waves. The CCES analysis description on OSF was registered in November 2021, after the 2020 fieldwork (https://osf.io/rqj8u).

2 Results

Across the writing and parking experiments, the pooled out-party effect is 0.12 standard deviations (SE = 0.04; 95% CI: 0.04 to 0.20). This is a modest shift relative to the variation in counts within each task. The pooled estimate summarizes these experiments; the individual contrasts show where the evidence is strongest. Capping all counts at 20 yields a similar pooled effect of 0.13 standard deviations (SE = 0.04; 95% CI: 0.05 to 0.21). Figure 1 shows

both specifications; caps of 10 and 15 also preserve the pooled direction (Appendix Table SI 4.1).

Across both writing studies, Democrats and Republicans find more errors when the text is attributed to the opposing party (Figure 1). These uncapped estimates all point toward partisan bias, although each confidence interval includes zero. The Lucid Republican estimate is sensitive to the right tail of the response distribution.

As Appendix Table SI 2.1 shows, Democrats find 8.7 mistakes when they think the text is written by a Democrat compared to 8.9 mistakes when they think the text is written by a Republican. On the other hand, Republicans find 7.6 mistakes on average when they think the text is written by a Democrat and 7.1 mistakes when they think it is written by a Republican. The estimated differences are small. The point is especially clear when you look at the medians, which are the same within each party. The confidence intervals for the differences include zero, but also leave room for larger effects (Appendix Table SI 2.5).

Appendix Table SI 2.2 presents the results of the replication on Lucid. Democrats find 5.5 (0.6) errors on average when they think Democrats wrote the text vs. 5.9 (0.8) when they think Republicans wrote it. On the Republican side, the corresponding numbers are 5.6 (0.9) (when they think Democrats wrote the text) vs. 4.9 (0.3) (when they think Republicans wrote it). Standard errors are in parentheses. These differences are imprecisely estimated (Appendix Table SI 2.5). Lucid includes 3 responses of 100 errors, while the group medians are four or five. Capping counts at 20 changes the Republican contrast from 0.71 errors (SE = 0.93) to -0.01 (SE = 0.39); the Democratic contrast changes from 0.43 (SE = 0.99) to 0.32 (SE = 0.39). Caps of 10 and 15 likewise leave the Republican contrast near zero or slightly negative (Appendix Figure SI 4.1).

Pooling the four writing contrasts in their original units yields 0.39 additional errors under an out-party cue (SE = 0.25; 95% CI: -0.09 to 0.87). For context, the in-party means

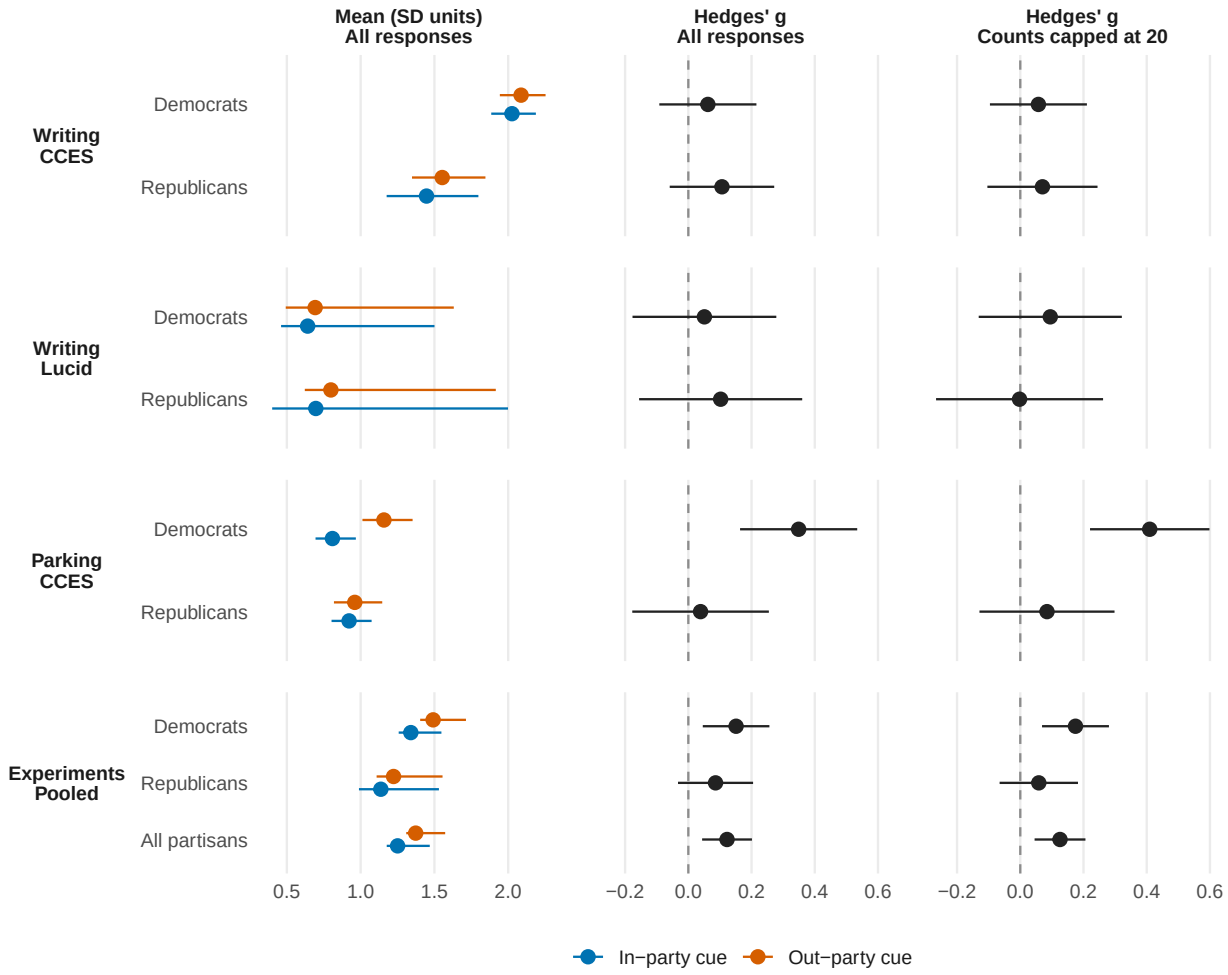


Figure 1: Party cues and reported counts among Democrats and Republicans, including leaners. Left: uncapped means in pooled within-condition SD units, with Hedges' correction. Middle: uncapped out-party minus in-party differences (Hedges' g). Right: the same contrasts after replacing every count above 20 with 20 and recomputing the SDs and synthesis weights. Positive differences favor the respondent's own party. The final row pools experiments by generalized least squares; the left and middle panels use the same weights. All intervals are 95% and use 5,000 respondent bootstrap draws, keeping each CCES respondent's outcomes together: percentile intervals for means and normal-approximation intervals for differences. CCES is unmatched and unweighted; Lucid excludes previews and retains consenting respondents who passed the attention check.

range from 4.9 to 8.7 errors. CCES supplies 87% of the inverse-variance weight. The pooled estimate is positive, but its interval still includes zero (Appendix Table SI 3.1).

We see a larger partisan bias in the experiment that manipulates the perceived party of the people who parked the cars. Democrats on average believe there are 6.1 poorly parked cars outside the Democratic Party meeting while they think there are 8.7 poorly parked cars in front of the Republican Party meeting (see Appendix Table SI 2.3). As above, the medians are much closer at 5.0 and 6.0 for the Democratic Party meeting and the Republican Party meeting, respectively. The difference is 2.63 cars (SE = 0.73) and remains positive when any single respondent is omitted. Switching to Republicans, the gap is much narrower—Republicans on average think that there are 8.4 poorly parked cars in front of the Democratic Party meeting and 8.1 in front of the Republican Party meeting. The mean difference is 0.34 cars (SE = 0.95); the gap in medians is one.

We also ask whether the effects are larger among strong partisans than among weak partisans and leaners. The estimated difference between these effects is 0.83 errors in CCES writing (SE = 0.55), compared with -0.63 in Lucid writing (SE = 1.33). In parking, the difference is 0.47 cars (SE = 1.20). These interaction estimates do not provide consistent evidence of larger effects among strong partisans (Appendix Table SI 3.3).

Turning to the results from the MTurk survey, Republicans count more people wearing masks than Democrats (see Appendix Table SI 2.4). We winsorized the responses at 15 because of extreme responses. Democrats on average spot 3.5 (0.1) masks and Republicans spot 4.4 (0.2) masks. The difference is 0.87 people (SE = 0.24), with a 95% confidence interval from 0.41 to 1.33. The medians are 2.0 for Democrats and 3.0 for Republicans. Republicans count more masked people in both survey waves and when responses are capped at 10 or 20 (Figure 2). Without a cap, reports as high as 80,000 dominate the averages and the estimated difference reverses sign.

The higher Republican counts indicate that Republicans report more mask-wearing among Trump supporters. This is consistent with a more favorable assessment of one's own side if mask-wearing is judged favorably; the survey did not measure that judgment. Unlike the text and parking experiments, this comparison does not randomly assign party cues. We therefore present it separately from the pooled experimental effect.

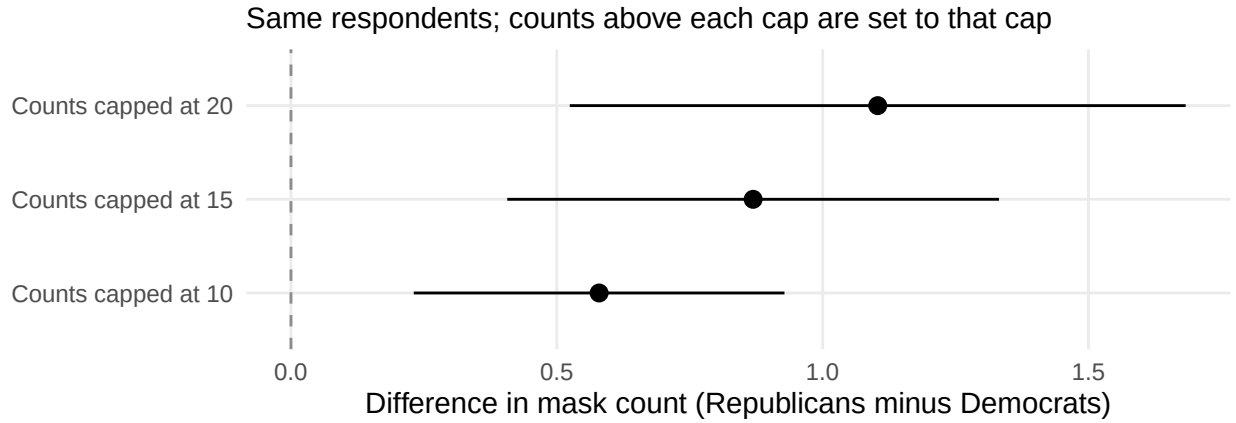


Figure 2: The mask-count gap under three response caps. Each row uses the same MTurk respondents viewing the same video of Trump supporters. Points show Republican minus Democratic mean counts after responses above the labeled cap are replaced by the cap. Intervals are Welch 95% confidence intervals. The sample pools both MTurk waves and retains respondents reporting humorous or insincere answers never or rarely. This is a descriptive comparison between respondent groups.

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SI 1 Treatments

Figure SI 1.1: Text treatment

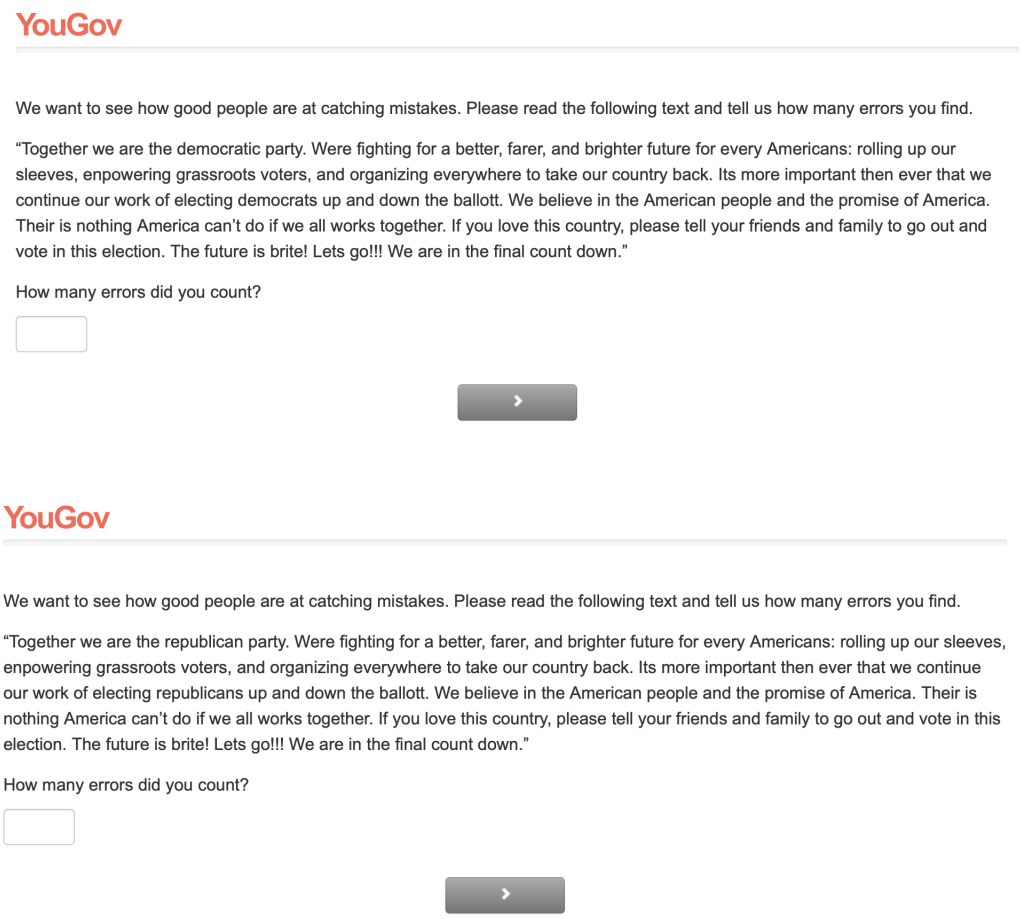
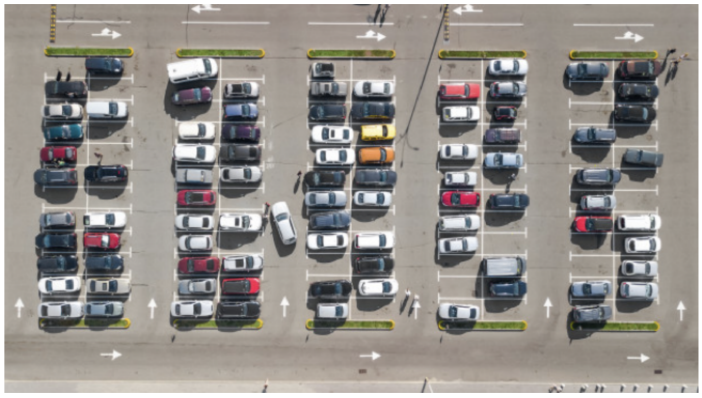


Figure SI 1.2: Parking lot

YouGov



We want to see how much attention people are paying in answering questions. How many cars would you say are poorly parked in this picture of the parking lot outside a Democratic Party meeting last year?

SI 2 Descriptive statistics and experimental contrasts

Tables SI 2.1–SI 2.4 report the descriptive statistics, including independents. Independents do not have an in-party or out-party cue; their means are reported by the party named in the treatment. Table SI 2.5 reports the experimental contrasts for Democrats and Republicans.

Table SI 2.1: CCES Writing

Respondents	Attributed party	N	Mean	Median	SE
Democrat	Democrat	334	8.65	9.00	0.23
Democrat	Republican	324	8.92	9.00	0.24
Independent	Democrat	94	9.07	9.00	0.72
Independent	Republican	110	8.24	8.00	0.43
Republican	Democrat	252	7.59	7.00	0.34
Republican	Republican	252	7.07	7.00	0.27

Table SI 2.2: Lucid Writing

Respondents	Attributed party	N	Mean	Median	SE
Democrat	Democrat	177	5.47	5.00	0.61
Democrat	Republican	132	5.91	5.00	0.78
Independent	Democrat	73	4.56	4.00	0.30
Independent	Republican	72	5.33	5.00	0.33
Republican	Democrat	111	5.58	4.00	0.89
Republican	Republican	109	4.86	5.00	0.26

Table SI 2.3: CCES Parking

Respondents	Attributed party	N	Mean	Median	SE
Democrat	Democrat	211	6.09	5.00	0.42
Democrat	Republican	212	8.72	6.00	0.60
Independent	Democrat	57	10.88	7.00	1.48
Independent	Republican	61	6.67	5.00	0.92
Republican	Democrat	181	8.43	6.00	0.63
Republican	Republican	163	8.09	5.00	0.71

Table SI 2.4: Mask counts capped at 15; never or rarely insincere

Respondents	N	Mean	Median	SE
Democrat	682	3.48	2.00	0.14
Independent	96	3.51	3.00	0.30
Republican	514	4.35	3.00	0.19

Table SI 2.5: Effect of party cues on reported counts. Differences are out-party minus in-party means within respondent party. Writing counts refer to errors in a passage; parking counts refer to poorly parked cars. Intervals are pointwise Welch 95% confidence intervals.

Study	Task	Respondents	Difference	95% CI
CCES	Parking	Democrat	2.63	[1.19, 4.06]
CCES	Parking	Republican	0.34	[-1.53, 2.21]
CCES	Writing	Democrat	0.26	[-0.39, 0.92]
CCES	Writing	Republican	0.52	[-0.33, 1.37]
Lucid	Writing	Democrat	0.43	[-1.52, 2.39]
Lucid	Writing	Republican	0.71	[-1.13, 2.56]

SI 3 Evidence synthesis and partisan strength

The writing synthesis uses inverse-variance weights to combine the four independent study-by-party mean differences in error counts. It assumes those differences are sufficiently comparable across the two passages to summarize in a common unit. We show the overall summary and summaries within respondent party (Table SI 3.1).

To combine writing with parking, we divide each out-party minus in-party difference by its pooled within-condition standard deviation and apply Hedges' small-sample correction. We use generalized least squares to summarize the standardized effects (Table SI 3.2). The sampling covariance comes from 5,000 bootstrap draws of respondents within study and respondent party. Each sampled CCES respondent contributes all available outcomes, preserving dependence between the writing and parking tasks. Covariances between independent study-by-party samples are set to zero. The scaled means in Figure 1 use the same standard deviations and Hedges corrections as the effects. Both cue means use the same synthesis weights, so their pooled difference equals the pooled Hedges' g. The scaled means use percentile bootstrap intervals to accommodate their skewed sampling distributions; the difference intervals use the normal approximation with the bootstrap covariance. The resulting estimate summarizes the observed experimental contrasts, rather than a random sample of possible tasks. The MTurk comparison is not included.

Strong partisans are CCES respondents coded 1 or 7 on `pid7` and Lucid respondents coded 1 or 10 on the provider's party-identification measure. Other partisans include weak identifiers and leaners. For each experiment, we fit a model fully interacting out-party assignment, strong identification, and Republican identification. We contrast strong and other partisans' treatment effects, giving Democrats and Republicans equal weight. Each experiment contributes one outcome per respondent; these interaction intervals use HC3 standard errors. Holm adjustment covers the three interaction tests. Table SI 3.3 reports both sub-

group effects and their difference, avoiding comparisons based on whether one subgroup's interval happens to exclude zero.

Table SI 3.1: Exploratory fixed-effect synthesis of writing contrasts. Differences are out-party minus in-party error counts. Inverse-variance weights combine independent study-by-party contrasts; intervals use the normal approximation.

Respondents	Difference	SE	95% CI	CCES weight (%)
All partisans	0.39	0.25	[-0.09, 0.87]	86.87
Democrat	0.28	0.32	[-0.34, 0.90]	89.92
Republican	0.55	0.39	[-0.22, 1.33]	82.11

Table SI 3.2: Exploratory synthesis of writing and parking experiments. Effects are Hedges' g, oriented as out-party minus in-party. Generalized least squares uses the covariance estimated from 5,000 respondent bootstrap draws within study and respondent party. Each CCES respondent's outcomes remain together in each draw. Intervals use the normal approximation.

Respondents	Pooled effect (SD)	SE	95% CI
All partisans	0.12	0.04	[0.04, 0.20]
Democrat	0.15	0.05	[0.05, 0.26]
Republican	0.09	0.06	[-0.03, 0.20]

Table SI 3.3: Exploratory moderation by partisan strength. Strong and Other columns report out-party minus in-party effects, with HC3 standard errors in parentheses. Other includes weak partisans and leaners. Each effect gives Democrats and Republicans equal weight. Difference is the strong-partisan effect minus the other-partisan effect, with a 95% interval. Holm adjustment covers the three interaction tests.

Study	Task	Strong	Other	Difference	95% CI	Adjusted p
CCES	Parking	1.71 (0.84)	1.24 (0.85)	0.47	[-1.89, 2.82]	1.00
CCES	Writing	0.79 (0.38)	-0.05 (0.40)	0.83	[-0.24, 1.91]	0.39
Lucid	Writing	0.27 (0.95)	0.90 (0.93)	-0.63	[-3.24, 1.98]	1.00

SI 4 Sensitivity to extreme responses

We repeat the experimental analysis with every count above 10, 15, or 20 replaced by the respective cap. These are sensitivity thresholds, not known upper bounds on valid answers. Each rule applies to all experimental groups and retains every respondent. For each cap, we recompute within-condition standard deviations, Hedges' g, the respondent bootstrap covariance, and the synthesis weights. These estimates describe differences in capped counts; their SD units differ from those of the uncapped estimates. Figure SI 4.1 shows all three caps alongside the uncapped contrasts, and Table SI 4.1 summarizes the pooled results.

Table SI 4.1: Pooled experimental effects under alternative response caps. Each cap applies to all study, task, and party groups. Standard deviations and synthesis weights are recomputed for each cap; all respondents remain in the analysis. Intervals use the respondent bootstrap covariance and normal approximation.

Response cap	Pooled effect (SD)	SE	95% CI
10	0.11	0.04	[0.03, 0.19]
15	0.12	0.04	[0.04, 0.20]
20	0.13	0.04	[0.05, 0.21]
Uncapped	0.12	0.04	[0.04, 0.20]

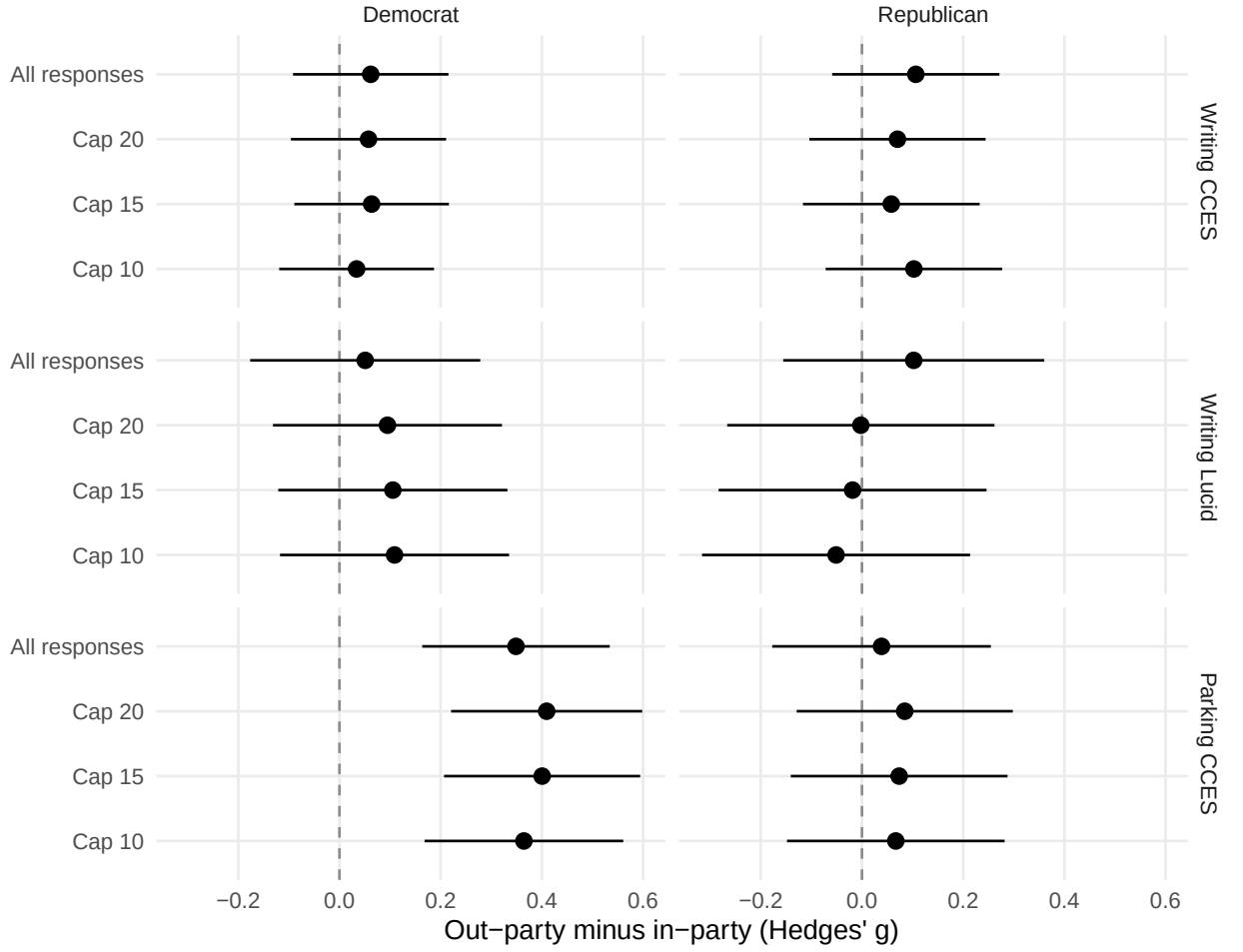


Figure SI 4.1: Experimental contrasts under alternative response caps. Points are out-party minus in-party Hedges' g among Democrats and Republicans, including leaners; intervals are normal-approximation 95% confidence intervals using 5,000 respondent bootstrap draws. Each cap replaces larger counts with the cap in every group. CCES respondents' paired outcomes remain together in each draw.